CampusWorld and BT's on-line education services

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CampusWorld was BT's first Internet service. Aimed at education users, it replaces an existing session-oriented bulletin board system that has been running for the last 16 years. This paper describes how the team turned the connectionless webserver technology into an access-controlled information service for the 1990s. It describes the hardware and network architecture that was used, the software that was required, problems that were encountered during development and how they were overcome. Finally it outlines how the CampusWorld platform has been used as the basis for a new set of products, CampusConnect, CampusHome and CampusProfessional.

1. What is CampusWorld?

CampusWorld [1] is an on-line educational service from BT. It replaces Campus 2000, a bulletin board style service with teletext style graphics which was introduced in the early 1980s. It was out of date in terms of its user interface and architecture and consequently was rapidly losing customers. Customers wanted access to the new Web technology with all that it offers, plus they wanted access to Internet mail to enable them to talk to educational establishments throughout the world.

CampusWorld is targeted at schools and colleges of further education and provides e-mail, file transfer and access to information databases such as FT Profile. The service is one of BT's on-line products and uses BT's access infrastructure — BTnet and Internet Dial Access. The service allows access to all the benefits of the Internet, the first real artefact of the much-heralded 'information age' and 'superhighway'. It allows schools to access the benefits already being discovered by commercial companies and universities — that they can discuss and collaborate in a range of ways, either one to one, via mail messages or one to-many via discussion groups. The benefits of the Internet can be tempered by some less attractive features, some of which have been explored in the media to the exclusion of all else. CampusWorld, and the other products in the Campus family, had tackled this problem by providing a feature rich 'walled garden' for schools to roam in, without allowing access to more undesirable information. The information within the 'walled garden' has been carefully checked for its suitability for schools. The schools can choose to keep within the walled garden protected area, or venture out (by special password) on to the full Internet.

CampusWorld went live on 8 September 1995 after three months of live trials with 20 schools situated all over the country.

The live service was an instant success and has now over 2500 schools and colleges using it. It offers these schools a number of facilities.

- Electronic mail

  Schools can communicate using electronic mail. Facilities allow for the creation of circulation lists, aliases and other Internet standard mail features. This allows schools to correspond with other schools or for pupils to individually correspond with 'pen-friends' across the world.

- Discussion groups

  Discussion groups can be set up so that topics can be open to a wide range of users.

- Internet access

  The service is split into a free corner, accessible by all, which gives a sample of CampusWorld features, and a 'walled garden' available only through subscription. Within the 'walled garden' schools have the capability to restrict access to undesirable sites or to allow access to the full Internet.

- School curricula on-line

  Curriculum material produced exclusively for CampusWorld by education specialists is available to be used by teachers in the classroom.

- Live news feed from an agency

  Schools have access to the latest news by a news feed supplied by UK News. This provides a wealth of current information on politics, human interest, sport, etc, which will build into a useful archive of information. The information can also be 'cut out' and used by schools to make their own newspapers.

- FT Profile gateway

  An optional feature that schools can request is access to the FT Profile service. This is an archive of news articles from all the major UK national newspapers and other sources, such as New Scientist. This can be reached using a simple-to-use text-based interface.

- Minitel

  Access to France Telecom Minitel service, using terminal software provided by a specialist French company. This enables schools to access the full range of its services including its telephone directory services.

2. CampusWorld development

2.1 Architecture

CampusWorld is based round World Wide Web (WWW) technology. Client software resides on a computer at the school or educational establishment and accesses the server application software using HTTP protocols. The server allows onward connection to the Internet through a proxy and access to X.25 services (Minitel and FT Profile) via a gateway (see Fig 1).

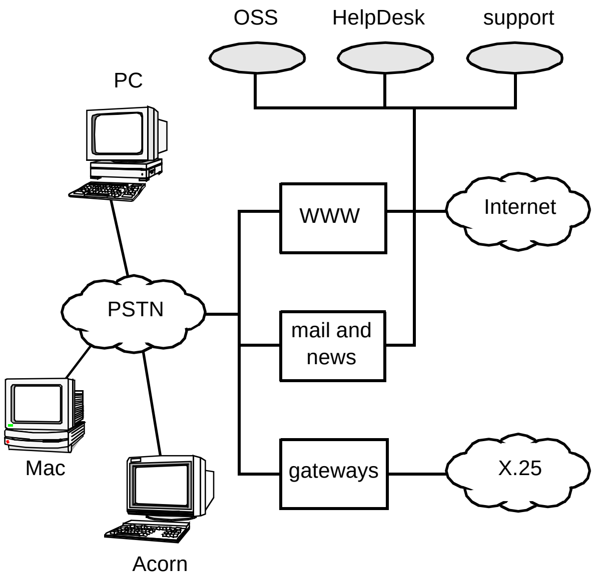

Fig 1 System architecture.

The service is currently run on three Unix boxes (all Sun Sparc 1000s with eight processors and mirrored disks). Currently one is used to support the Web servers, another for the X.25 gateways (and other communications), with the third supporting Mail and Usenet News. This initial configuration is under review and is soon to be changed now that there has been time to analyse the load data. All the machines run SQL*Net which gives them access to the customer database. A news feed for Usenet News is supplied by BTnet, but most news groups are currently filtered out because they carry unsuitable material.

2.2 CampusWorld administration

When considering the administration of CampusWorld it was apparent that with the large numbers of changes occurring in a school's environment there could be a huge operational burden placed on BT which would be difficult to cost justify. To this end an administration model (see Fig 2) was developed that allowed BT to create the 'initial' supervisor/administrator account, for the customer to control the individual user accounts and for a help desk to support customers with queries. The school's administrator (typically one of the teachers) can create new accounts for individual pupils, assign permissions such as Internet access to the user, to upload content and to administer passwords. Another key decision taken in the design is that as all of the administration functions (except system administration) were to be accessible via a standard Web browser, there is no special client software to install and training of support personnel and customers is minimised. The server applications to support the administration use specially protected CGI (common gateway interface) scripts. This makes adding users or modifying user details very easy for the helpdesk staff, and no specialist knowledge of Unix is required. Third-line support can access using Telnet login but only from specified IP addresses within BT's firewall. This provides added security.

The overall result has proved to be very successful, the customers do indeed manage their own accounts with little or no involvement from BT. The lessons learned from this approach are now being used in BT's new intranet platform where a similar administration model will allow companies to manage their own intranet on a BT-hosted platform.

2.3 Client

Customers ordering the service (at a basic cost of £12 per month) receive user-guides and software for either PC, Mac or Acorn. This support makes CampusWorld unique in offering a service to all types of computers found in mainstream UK education.

Included in the software pack is a TCP/IP software stack that enables the 'PC' to talk via a modem (or LAN) to the CampusWorld computer. A Netscape Web browser is included as well as a Minitel/FT Profile helper application.

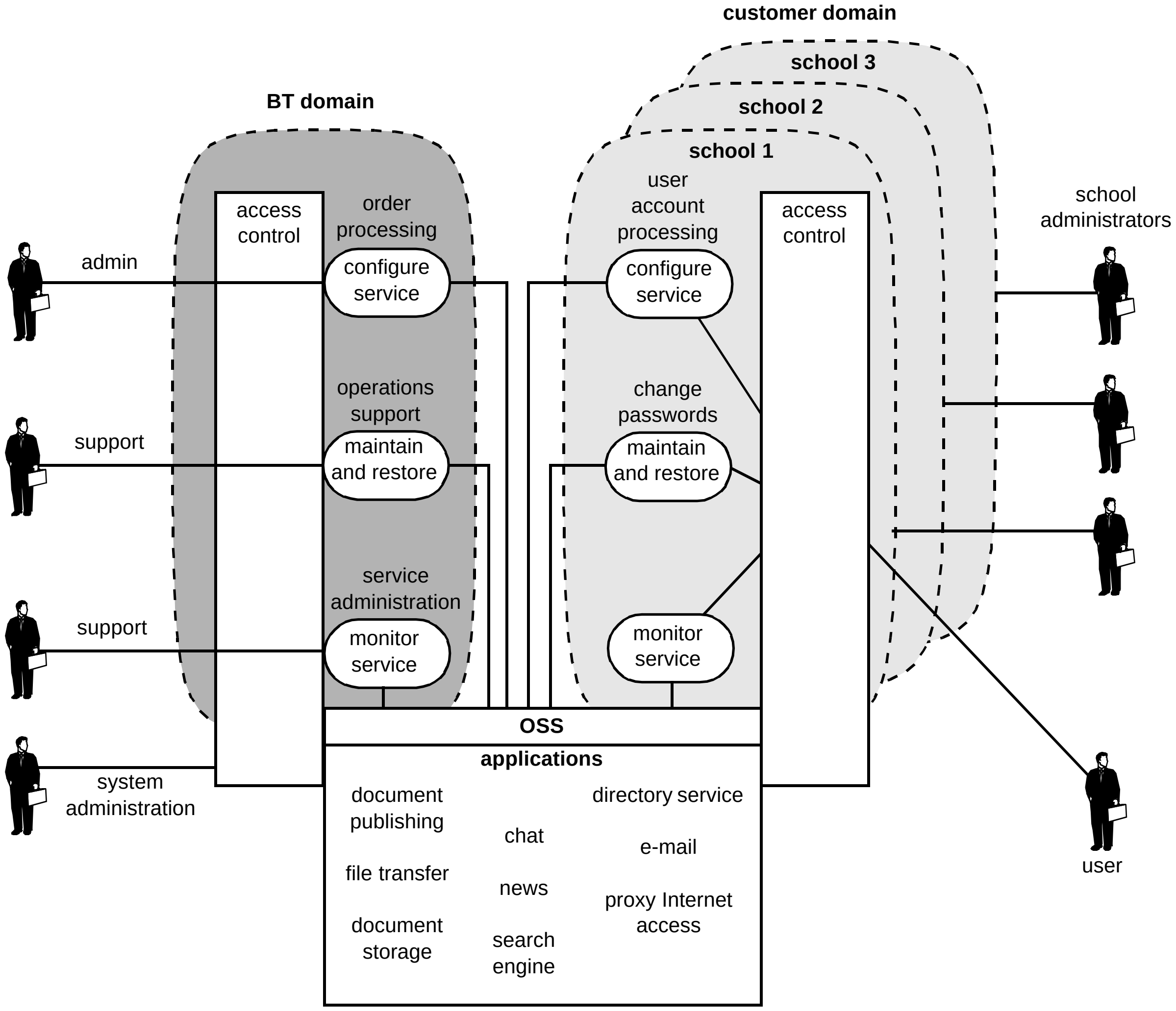

Fig 2 Administration model.

To use the service, schools need only have a computer (PC, Macintosh or Acorn) and a modem. For more sophisticated set-ups the schools can access the service using ISDN. In this case the ISDN lines need to be installed and LAN systems set up. The client software can gain access to CampusWorld using one of the following three methods.

- Over the public switched telephone network (PSTN)

  This uses a standard telephone line. To send data over this line you need to have a modem at the customer's end. BT provides the connection point to Campus-World, also called the 'point of presence' (PoP). This is achieved using a PoP in London and an 0345 number to provide local rates for all users. This type of access is restricted to single computer use. With this type of connection the speeds or access can now be as high as 33.6 kbit/s.

- Over the integrated services digital network (ISDN 2)

  This type of connection can transfer data at up to 64 kbit/s. Telephone lines with special installation are required, but these provide much faster data access than PSTN. It also becomes feasible to network computers since a LAN will allow several computer terminals to access CampusWorld at the same time.

- Using other Internet service providers

  Schools which already have an account with another Internet provider can access the public area of CampusWorld, although to explore the information within the 'walled garden' users need a user ID and password which is available only by subscription to the service.

2.4 *New authentication client*

The enhanced client provides a separate log-in program which ensures that all user log-in information is encrypted between the client and the server. This makes the system significantly more secure while offering a number of future options that can be exploited. It also improves the access speed issue which was partially solved by caching because now the access rights of a user are only checked on the first log-in rather than with each page access, as is normal with webservers.

This new software offers a number of potentially exciting new features which may be taken up in later releases.

- Automatic update of user configuration files

  When a user connects to the system version information is exchanged with the server. This offers the option to download the latest version of the files either automatically or at the user's behest.

- Ability to limit the number of simultaneous sessions

  Because the log-in method establishes the concept of a connected session, it is possible to limit the number of allowed concurrent sessions that a user or organisation is allowed. This could prove especially useful, enabling the charging of an appropriate fee to an organisation with many users connecting via a wide-bandwidth network.

- Secure file transfer

  Any files transferred either from the client or to the client are encoded using BT strong encryption technology. This overcomes one of the main problems of the Internet — that of poor security caused by the use of US export-quality encryption methods.

- Secure financial transactions

  The transmission of a user's PIN with ordering information, as in the previous point, is encrypted using BT strong encryption techniques. It is essential for the development of commercial transactions on the Internet that PINs can be kept confidential, to overcome customer fears of fraud.

2.5 Server

Server software is based around the popular CERN web server which has been modified to improve its performance and add additional security features.

The system uses an Oracle database which stores information on customer account details, ordering information for CSS, and user profile information. One important design principle was that the schools should be able to administer their own accounts. This gives them the flexibility they require to use the service as they want, and in turn reduces the load on BT in managing the service. As described above, when a school is added to the service they are given a supervisory account which can be set up to assign access rights to sub accounts for that organisation. They can also purchase a number of mailboxes (one is included in the basic package) and assign it to one or many accounts.

System functions are provided through CGI programs. These can read, write and update the database and enable the server to provide facilities such as directory searches and modification of account details.

The server makes use of a number of packages, commercial and public domain, to provide CampusWorld facilities. A Glimpse search engine is used to allow searching of the Web pages, and the Nexor XTPP mail server provides SMTP-based mail and potential X.400 and facsimile gateways. The standard Usenet News daemon was used to provide the discussion groups, with CGI to allow easy configuration. Recently the discussion group feature has been provided by using a commercially available product called Caucus.

2.6 Content management

The schools access the service to use both the facilities it offers and the content it can deliver. The user interface has been designed to be used either by teachers or pupils, so simple clear graphics coupled with a button bar were incorporated, to ease navigation around the system.

In CampusWorld this content is generated by curriculum specialists hired directly by the product team. To load the content on the live machine requires a separate content delivery machine which is outside BT's internal firewall. Content is automatically checked by a number of programs and is then transferred to an intermediate, or staging machine, inside the firewall. It is only from this machine that content — under the control of the product team — can be loaded on to the live system. The staging machine is also used to test new software releases.

Not all content comes through this route; some content is automatically loaded from file servers and may also need to be converted from its format into HTML. One example of this is the UK news feed.

2.7 Problems and solutions

There have been a number of problems associated with the release of CampusWorld — mainly caused by its sheer success. Currently it is taking over 100 000 hits per day, mainly between 8 am and 5 pm.

The first problem encountered was not being able to take orders fast enough. This meant that a streamlined version of the administration software (which is also WWW based) had to be produced.

The second major problem was that when the system reached around a 1000 users the access control method used in the CERN web server became overstretched. To get round this problem the access control modules had to be modified so that improved caching was used to speed access to user information. This produced an immediate ten fold speed-up on using the service.

Another problem encountered was that as the number of accounts grew any drop-down menus provided quickly increased to an unmanageable length. This is because the HTML page has to download the entire contents of the drop-down menu before it can display the page, and as the size grew to several thousand entries the page size increased. Also some PCs and Macs simply crashed, unable to cope with menus of that size. To overcome this, search criteria were added to the menus to ensure only a few entries appeared in the menus.

3. The future

CampusWorld was BT's first on-line Internet service developed in a very short time-scale. Since its launch in September 1995, ways have been explored in which the service can be extended and also how similar benefits can be provided to others in the education sector.

In addition to the authentication client, it is anticipated that features developed in BT will be added (currently CampusWorld supports the network summariser, a BT Laboratories development that enables text-based Web pages to be 'summarised' using artificial intelligence to reduce documents by up to 80% in size). Further developments are likely to see such things as text-to-speech programs, audio and video streaming, language tutors and talking head.

Using CampusWorld as the 'base' product and maximising the reuse of core code, the Campus product family has now been extended to include CampusConnect and HomeCampus. These three services will soon be joined by a fourth, CampusProfessional; all are discussed in more detail below.

3.1 CampusConnect

CampusConnect [2] is another educational service but is targeted at school administrators and bursars. This service launched in July 1996 enables the school to send examination entries, receive examination results from the Examination Boards and also exchange information with their LEA all in a fully encrypted secure manner. Further, in CampusConnect an additional security feature was added which enabled the examination result files to be doubly encrypted. The second level of decryption was achieved by the use of a special decryption phrase which the users were given at a pre-defined time. This phrase enabled them to 'unlock' or decrypt their results files. By using this delayed decryption, examination result files were able to be delivered to the schools (examination centres) up to 48 hours prior to the time of official publication. All BT did was to publish the decryption phrase ('open sesame' or similar) at the predefined time. This cured the excessive simultaneous access requirements of the old system and added an extra level of security. This delayed decryption feature is available to all Connect administrators. This service has built on the security features within CampusWorld adding secure file transfer and bespoke mail. It will also provide a shopping mall for schools to purchase goods and an information service for central bodies such as the Department for Education to disseminate information to schools. Very clearly CampusConnect is an intranet; it enables closed user groups of communities of interest, such as LEAs, to share information among their closed community with complete confidence in the information integrity and security.

3.2 HomeCampus

CampusWorld was initially targeted at schools but a residential service is now available. HomeCampus [3] was launched in December 1996 and offers some of the CampusWorld functionality along with additional more 'fun' material suitable for the home user.

The home system produces several interesting design challenges. Firstly, the number of potential customers was much larger than with the academic service. Secondly, the customers required on-line registration with payment by credit card. Lastly the product had to be simple to use and provide controlled access to the less desirable aspects of the Internet — all of which have been achieved.

3.3 CampusProfessional

CampusProfessional is an on-line training service aimed at meeting the training needs of the corporate and small and medium enterprises.

It will bring CD Rom type material to the students' desk both at work and at home. To assist the student in selecting course material the service content will be arranged in academies — management, technology, finance and so on.

Further, by a series of intelligent questions and answers users will be able to navigate to the content that most meets their learning style and current ability or competence in that subject. BT aims to trial this product in the summer of 1997.

4. Conclusions

The Campus products are a considerable success. They are based on standard Web technology wherever possible, which has enabled a very rapid deployment. The development teams are adding innovative enhancements to the standard features to give Campus customers the best possible products. In the future these innovations will be extended in some exciting new areas and also rolled out to a family of education products — and possibly beyond. The front Web page of Campus can be located at URL http://www.campus.bt.com/

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References

1 CampusWorld product definition, CAMPUS:TECH:SPEC:006.

2 CampusConnect product definition, CAMPUS:TECH:SPEC:007.

3 CampusHome product definition, CAMPUS:TECH:SPEC:015.

Phil Leveridge joined BT as a student in 1980. In 1981 he went to Birmingham University to study Physics. After obtaining his degree he joined BT Laboratories where he initially worked on a 565 Mbit/s optical fibre communications system. In 1986 he went to Essex University to study for an MSc in Telecommunications.

After obtaining his MSc he returned to Martlesham to work on ‘Work Manager’, a work allocation system for field engineers. He then became the project leader for an automated directory enquiry system using speech recognition. For the past year he has been the project leader for server-based application development on CampusWorld and now CampusConnect Internet projects.